

* CGM *

(Computer Graphics & Multimedia)

Computer graphics is the process of creating, displaying, manipulating & storing pictures/images using a computer.

Ex Games, Movies, 3D models, Maps, graphs, GUI, Simulations, Animation etc.

- > It is used for displaying information in the form of pictures, graphs & charts.
- > Objects are presented as a collection of discrete picture element.

Pixel is smallest screen element.

- > Computer graphic device is combination of some input devices & display devices on which user can edit graphics using keyboard, mouse & sensitive touch panel.

Advantages of Computer graphics

1. It provides the facility of high quality graphics on computer screen.
2. It provides various tools to create & ~~multimedia~~ manipulate pictures easily.
3. Computer graphics can be used to produce animations & dynamic movement of object.
4. Properties of an object can be changed using update tools of objects.

5.

* Application of Computer graphics

→ Computer Aided design (CAD) to design & represent objects & structure on computer.

Ex: Building & structure design, manufacturing, Automobiles, Aircraft, Ships, Computer chips etc.

→ Medical Applications to design 2D Coloured images representing cross-section of human body or organ.

Ex: medical imaging, Human body visualization, organ visualization etc.

→ Entertainment like movies, videos, cartoons, Animation, video games.

→ Infographics, fashion art to create digital artwork & illustration.

→ Simulation & virtual reality to make simulate of complex processes.

Ex: Flight simulation, mobile simulation etc.

→ Desktop App publishing like Journal, newspaper, newsletters, Books, Publishing material.

* Interactive & Non-Interactive Computer Graphics

1) Interactive Graphics

In which displayed image is under the control of the user.

Ex: Computer games. (According to user input, graphics move or created.)

3D model software, interactive maps etc.

2) Non-interactive Graphics

In which displayed image is not under the direct control of the user.

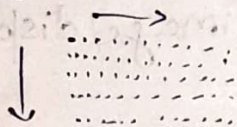
Ex: Screen savers, movies.

→ No direct user input.

→ follows predefined sequence.

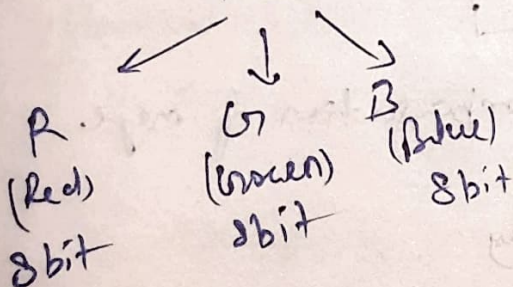
* Pixel Smallest element of digital image that contain information about image.

arranged in Rows & columns in 2D.



Intensity of each pixel is variable (brightness)

Pixel have 3 Component of basic colors.



** Total 24 bit or 3 byte in 1 pixel

$$\text{Shades} = 2^8 \times 3$$

$$= 16.7 \text{ million}$$

- Image Size : How many pixels in a image is size.
Total pixel = Height $\times$ Width.

Storage Size $\Rightarrow$ ~~Total~~
Total pixel $\times$ Bytes/pixel
(3)

Image Size in byte = $\frac{H \times W \times B}{8}$ ($B = \text{bit/pixel} = 3$)

- Image Resolution It refers to the number of pixel used to represent an image, generally expressed in $W \times H$.

- Pixel Density Number of pixels represent per inch of a display.
PPI = Pixel per inch.

PPI = $\frac{\sqrt{W^2 + H^2}}{D} = \frac{\text{Diagonal pixel}}{\text{Diagonal inches}}$

- Aspect Ratio Ratio of width of an image/display to its height.

$$\text{Aspect Ratio} = \frac{\text{width}}{\text{Height}}$$

$\rightarrow$ used to understand shape & propagation of image.

Common Ratios

4:3 $\Rightarrow$ older standard display

16:9 $\Rightarrow$ widescreen ratio

SD - Standard definition = 4:3

HD - High definition = 16:9

Full HD - 16:9 (1080p)

2K to 4K resolution, $\approx$ 16:9

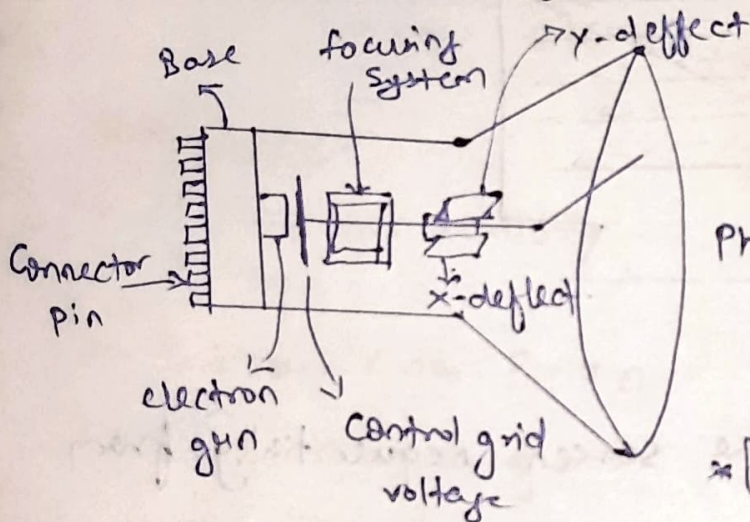
* Display devices in Computer Graphics

→ Output device used to represent the information in the form of image (visual form)

→ Display system (video monitor) or VDU (Video display unit)

Ex = CRT, Color CRT, LCD, LED, DVSIT, 3-D display etc.

1) CRT (Cathode ray tube): - It converts electrical signals into visual information by controlling an electron beam to produce images on a phosphor-coated screen.



* [Focusing System (Cathode)
Base (Cathode)]

Components

(i) Electron gun produces electron beam.

In this it contains, Heater, cathode, control grid & Anode.

(ii) Heater Heats cathode & cathode emits e-
Control grid controls the beam intensity.

• Anode give direction to e^- .

→ Phosphor screen CRT inner screen is coated by phosphor.

High speed e^- falls on this screen that create light.

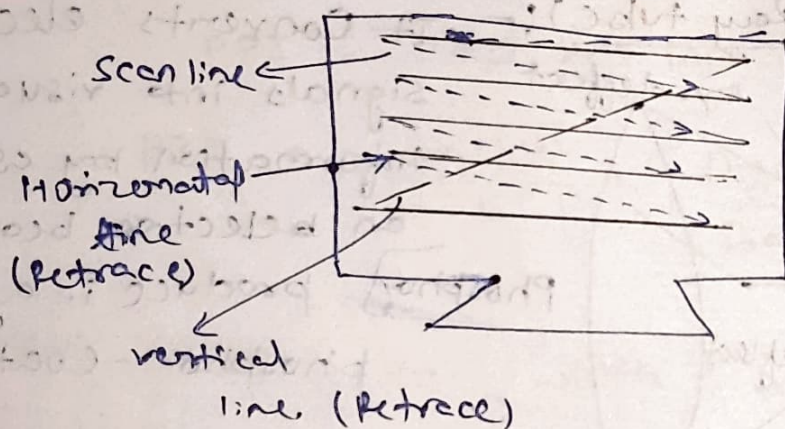
* There are two major CRT Scanning Techniques.

(i) Raster scan

(ii) Random scan.

(a) Raster Scan

A display technique in which the e^- beam scans the entire screen from Top to bottom, one horizontal line at a time, to produce an image.



→ Electron beam scan the screen sequentially from top to bottom.

→ It have two retrace

Horizontal retrace movement of the e^- beam from right end to left end of line

Vertical Retrace Diagonal line Bottom right to top left.

→ Refreshing is the process of repeatedly scanning & redrawing the image on the screen to maintain its visibility.

It create 50 to 60 per refresh per sec.

• Refresh Buffer (Frame Buffer)

Memory area used to store the intensity or color information of the pixels that make up the image.

Note: Bitmap (Black & white screen)

0 → Black

1 → White

then frame buffer is known as bitmap.

• Pixmap for Colorful image.

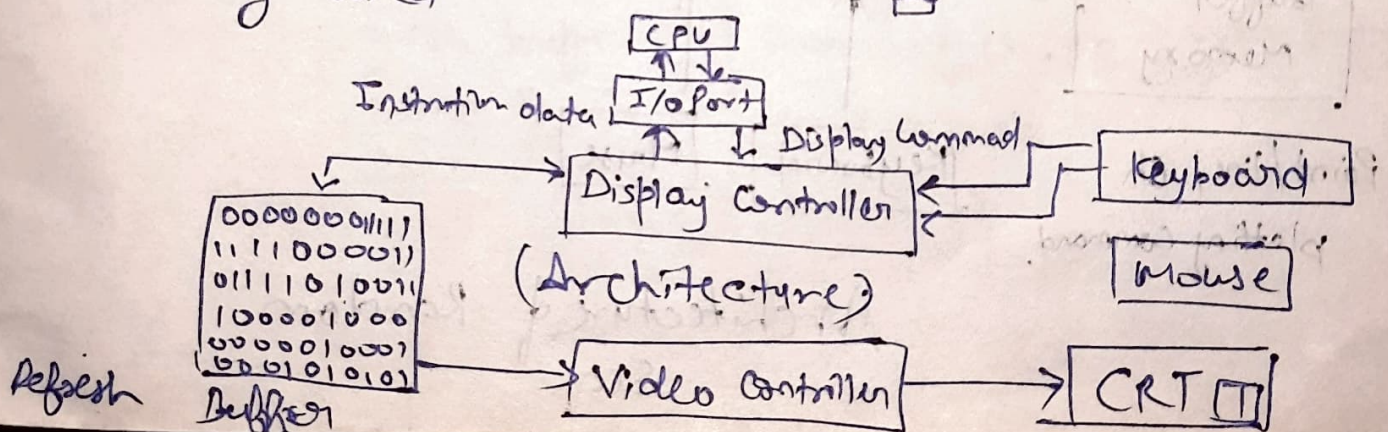
where 24 bit/pixel.

Advantage

- Suitable for realistic image
- Support Color Pattern
- Large number of color
- widely used

Disadvantage

- Lower line quality
- Aliasing / zig-zag effect (discrete pixels)

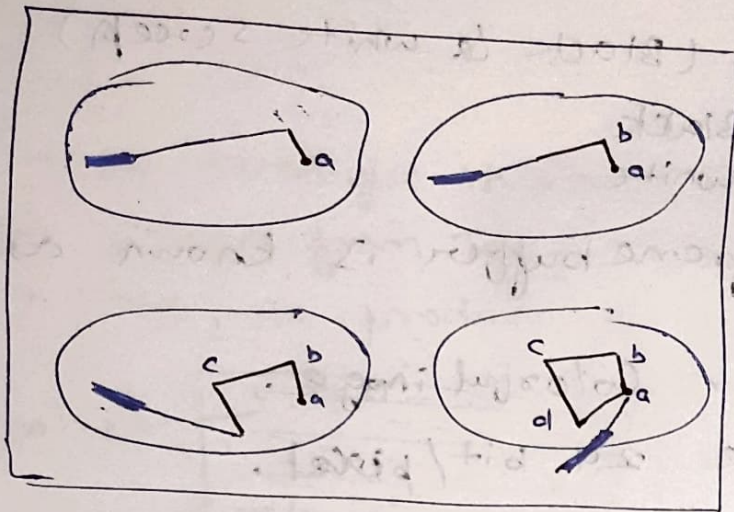


(b) Random Scan :- A display technique, in which the electron beam is directed only to the required positions on the screen to draw the image. Also known as

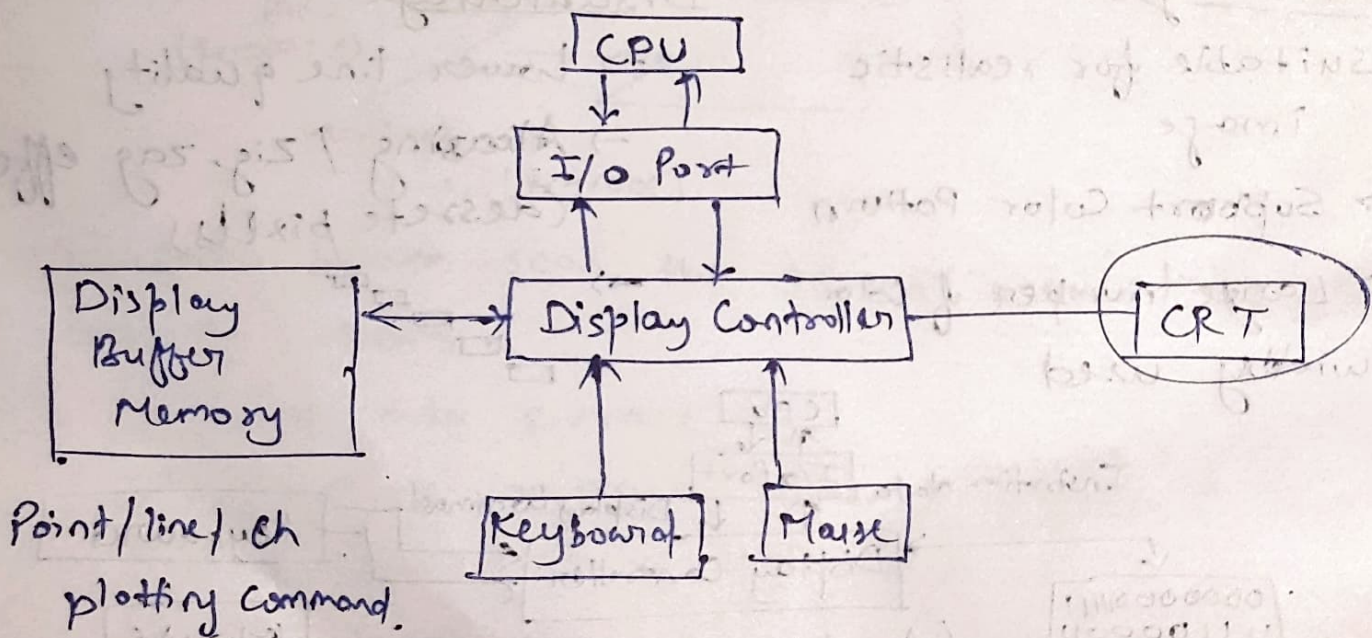
Vector Scan or Stroke writing or Calligraphic

→ It doesn't scan every pixel of entire screen.

→ Beam can be directed to different ~~dis~~ required location according to instruction.



along a vector of direction.



Architecture of Random Scan

It has Display Buffer

It stores instructions required to draw the image.

Instruction (display list)

Ex: Move To A

Draw line to B etc.

Controller interprets these display list & directs the electron beam accordingly.

→ It also requires refresh. So repeatedly executes the display list / required vectors.

Advantage

- Smooth lines
- High resolution
- Suitable for line drawing

Disadvantage

- Not suitable for realistic image
- limited color/shading capability
- Expensive & refresh depends on number of lines

Difference between Random & Raster Scan

<u>Raster Scan</u>	<u>Random Scan</u>
1. Collection of pixels	Composed of path & points.
2. scan conversion is required	scan conversion isn't required
3. Less Costly & cheap	Highly costly
4. Frame/Refresh buffer	Scan buffer. (Display Buffer)
5. Large number of colors	limited colors
6. low resolution	High resolution.
7. Realistic images	lines & shapes
8. Refresh whole screen	Refresh only displayed area
Ex: .jpg, .png, .jpeg etc	Ex: .svg, .wmf etc

* Graphics Input Devices :-

Graphic input devices are hardware device used to provide graphical information, commands, coordinates or user input to a computer graphics system.

Ex Keyboard, Mouse, Trackball, Joystick, LightPen, Graphics Tablet, Touch Panel, Image Scanner

* Keyboard

used to enter text, number, commands & control instructions into a computer.

used for -

- enter commands
- enter coordinates
- set object properties
- select menu/options.

* Mouse

A pointing input device used to select, position & manipulate object on the screen.

used for -

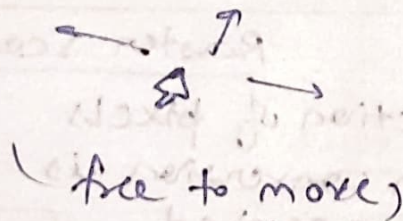
pointing

selecting

Clicking

Dragging

Drawing



* Trackball

A stationary pointing device

user rotate a ball. → cursor moves

Requires less physical space than mouse.

Uses CAD
Graphics System

* Joystick Mainly controlling the direction & movement of object.

→ A stick moves in different direction.

Application

→ Computer games

→ Flight simulation

→ Vehicle simulation.

* Light Pen Light pen is a pen-shaped input device used to select or draw directly on a display screen.

(Pointing device)

(digitizer)

* Graphics Tablet A digitizer converts physical drawing or movement into digital coordinates. It has a flat tablet & stylus/pointer.

Uses → CAD

- Engineering Drawing

- Digital Art

- Handwriting input.

* Image Scanner -

Scanner converts a physical image/document into a digital representation.

Useful For importing photographs, drawing & documents into a graphics system.

* Graphics & Software & Standards :-

Graphics software is a set of programs used to create, manipulate, store, display & manage graphical images & objects using a computer.

Eg - CAD Software

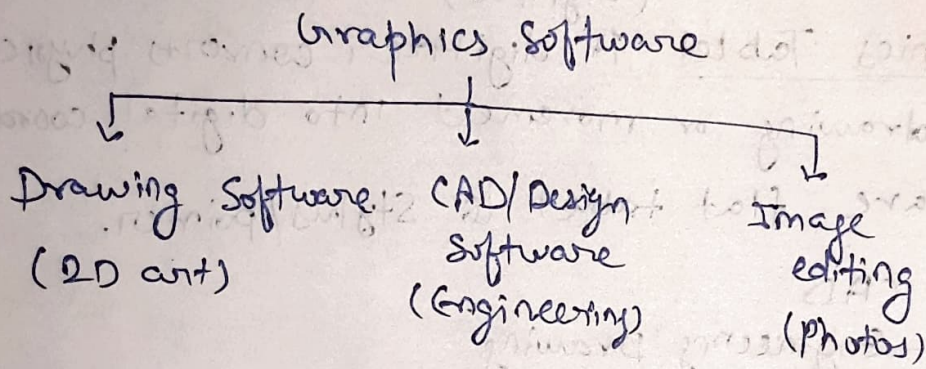
- Drawing Software

- Image editing Software

- Animation software

- 3-D modeling software

Types of Graphics Software



(a) Drawing Software (Painting)

Used to create & edit 2D graphics

Eg Digital drawings

Illustration

Poster

logos

Basic operation

Draw

Erase

Fill

Resize

Rotate & change color.

(b) CAD Software

(Computer Aided design);

Used to create technical & engineering designs.

Ex mechanical design

Building design

Automobile design

Circuit design.

CAD is particularly important because computer graphics provides precise representation of geometric objects.

(c) Image Editing Software

Used to modify existing image.

Operations including -

cropping

Resizing

Colour adjustment

Brightness adjustment

Applying effect

Removing / adding objects.

Ex Canva

Photoshop etc.

Figma etc.

(d) Animation Software

Used to create moving graphical objects.

Ex

frame 1 $\rightarrow$ frame 2 $\rightarrow$ frame 3 $\rightarrow$ frame 4
motion.

Animation is essentially the display of a sequence of images/frames to create the appearance of movement.

(c) 3-D graphics Software

Used to create & manipulate three-dimensional objects.

Typical operations -

modelling

Rotation

Scaling

Translation

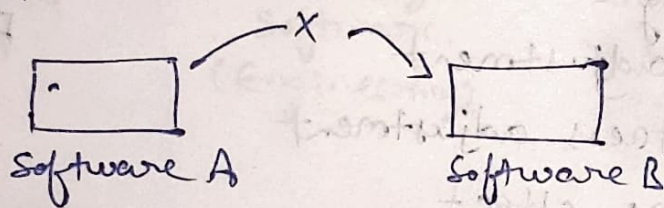
Lighting

Rendering.

* Graphics Standards:-

A standard is set of rules/specification that defines how graphics system, software or devices should communicate & represent graphical information.

need?



If both use completely different methods of representing graphics, exchanging data becomes difficult.

Purpose maintainability, compatibility, portability & interoperability, consistency.

Some graphic standards are

GKS

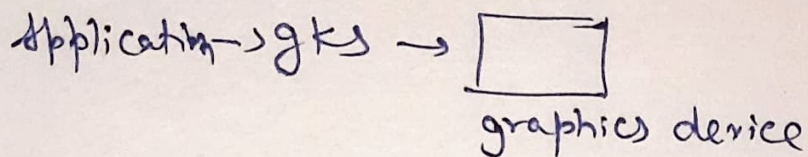
& PHIGS

* GKS (Graphical Kernel System)

It is a standardized graphics system/interface for creating & manipulating graphical information.

→ It provides a common set of graphics functions, so applications don't have to ~~spend~~ depend completely on a particular hardware device.

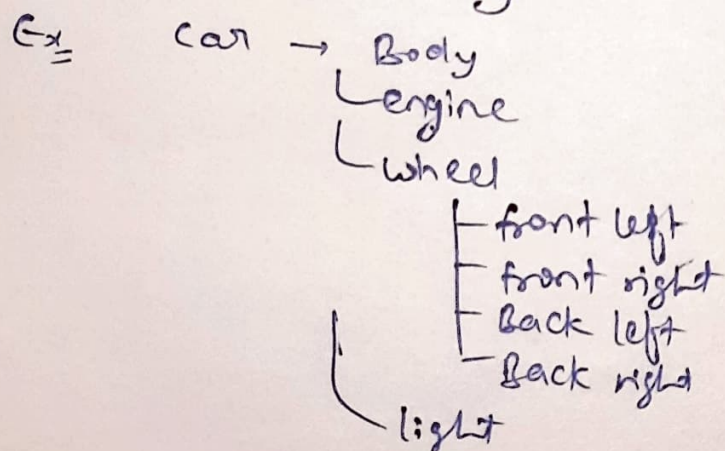
→ It acts as an interface between the application & graphics device.



* PHIGS (Programmer's Hierarchical Interactive Graphics System.)

It is a graphics standard designed especially for interactive graphics & hierarchical structures.

→ It allows graphical objects to be organized in a hierarchy.



→ useful for complex graphical models.